

PETsys
Electronics

TOFPET2 ASIC Evaluation kit

Hardware User Guide (v1.5)

Rev. 1.5: May, 2019

Revision history

The following table shows the revision history for this document.

Date	Rev.	Description
xx/05/2017	1.0	Initial release – Evaluation kit with TOFPET2 ASIC
xx/06/2017	1.1	Replacement of sentence on page 18 d), and page 19 c)
xx/05/2018	1.2	Added a new table
xx/09/2018	1.3	Front-End-Module (FEM) FEB/D_v2-1Kv2 motherboard related: - PW1..8 connectors - P2 polarity BIAS-16P Mezzanine
xx/12/2018	1.4	Default HQCD cable length changed to 50 cm New peltier p/n Disclaimer
xx/05/2019	1.5	Peltier control revised (cooling section)

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1 Safety warnings

- Never plug or unplug the SiPM¹ while the system is powered
- Be very careful with the polarity of the SiPM when plugging it
- Do not expose the SiPM to a large amount of light while it is biased
- Keep the temperature of the SiPM controlled
- Typical ESD² precautions like having both the workbench and the wrist connected to a good electrical earth should be taken in order to prevent abnormal failures, malfunctions or even permanent damages.

¹ Silicon Photo Multiplier

² Electrostatic Discharge

2 Product Information

The PETsys TOF ASIC Evaluation kit (e-kit) is a complete SiPM readout kit that can be used in applications where only a small number of channels need to be read, and it provides an easy to use low cost way for exploring the unique, powerful and versatile features of the PETsys TOFPET2 ASIC.

The standard version of the e-kit comes with one FEB/D_v2 board, one GbE ethernet link, and two basic Front-End-Modules ([FEM](#)), that is, each FEM has only one 64-ch. High performance TOFPET2 ASIC, FEM-64 in short, whereas the standard module is made up of two 64-ch TOFPET2 ASICs, FEM-128 in short. Depending on the FEM type, each can read directly either one or two 8x8 SiPM arrays compatible with KETEK (e.g. p/n PA3325-WB-B0-A0-0808) and HAMAMATSU (e.g. p/n S13361-3050AE-08) manufacturers, but with some small adapter boards, these will equally well read analog SiPMs from other vendors.

Flexible cables connect the FEMs to the FEB/D board, providing great flexibility in the experiment setup.

We provide all the firmware and software needed for acquiring and analysing the signals of the TOFPET2 ASICs. The source code of the DAQ software in the DAQ computer is provided, giving the user all freedom to explore the performance of the system, or to adapt it to the user particular needs.

For an easy start we supply the e-kit with two KETEK SiPM pixels mounted on a board that plugs in the FEM, two LYSO crystals of 3x3x5 mm², and a holder for placing a standard circular ²²Na source in-between. We also provide a cooling box with Peltier elements, which combined with an external sort of temperature regulation, either manual or automatic (see please page 22 to get further details), allows stable and reproducible measurements. The box also protects the measurement setup from ambient light.

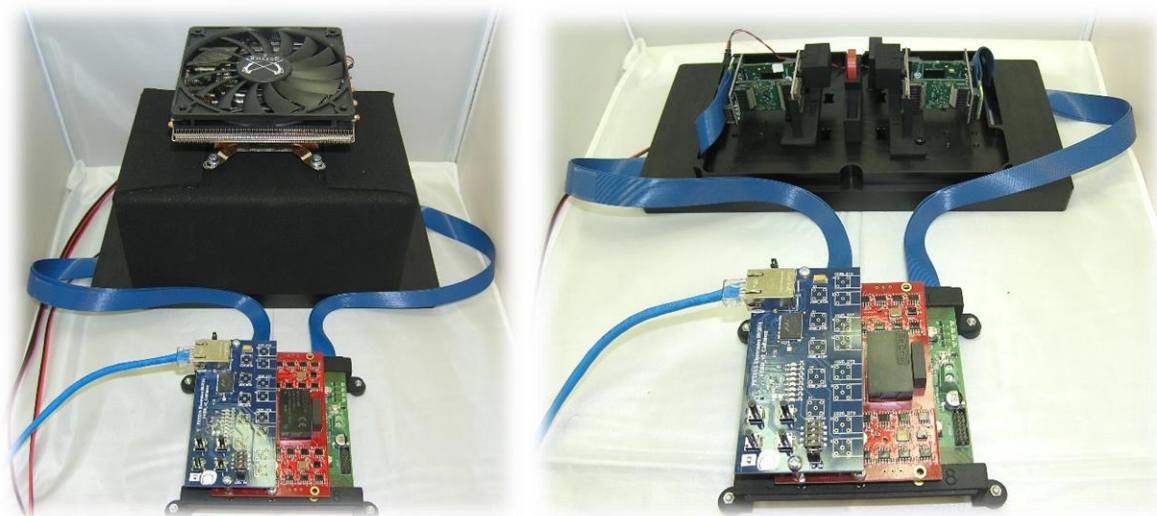


Figure 1 – Complete test setup with and without cooling box

3 Main features

- Two FEM-64 and two cables for connecting them to the FEB/D_v2 board.
- Two SiPM pixels (KETEK 3x3mm) mounted with two LYSO crystals and a standard circular radioactive point source holder (source not included).
- A cooling box with Peltier elements to have stable and reproducible measurements.
- Supplied with pre-installed firmware on the FEB/D_v2 for configuration of the ASICs and readout of the data using Gigabit Ethernet.
- Application software for receiving the data in a PC.
- On board DC-DC converters supply power to the ASICs in the test boards
- SiPM bias voltages produced in a mezzanine board. The default mezzanine provides 16 positive bias lines, 0-75 V, 20 mA peak or 2.5 mA average per bias line
- Only requires a single external supply voltage of 12 Vdc/4 A and one computer running Linux RHEL/CentOS 6.x or 7.x, 64-bit.
- Optionally can accept a veto signal

4 Hardware

4.1 FEB/D_v2 set

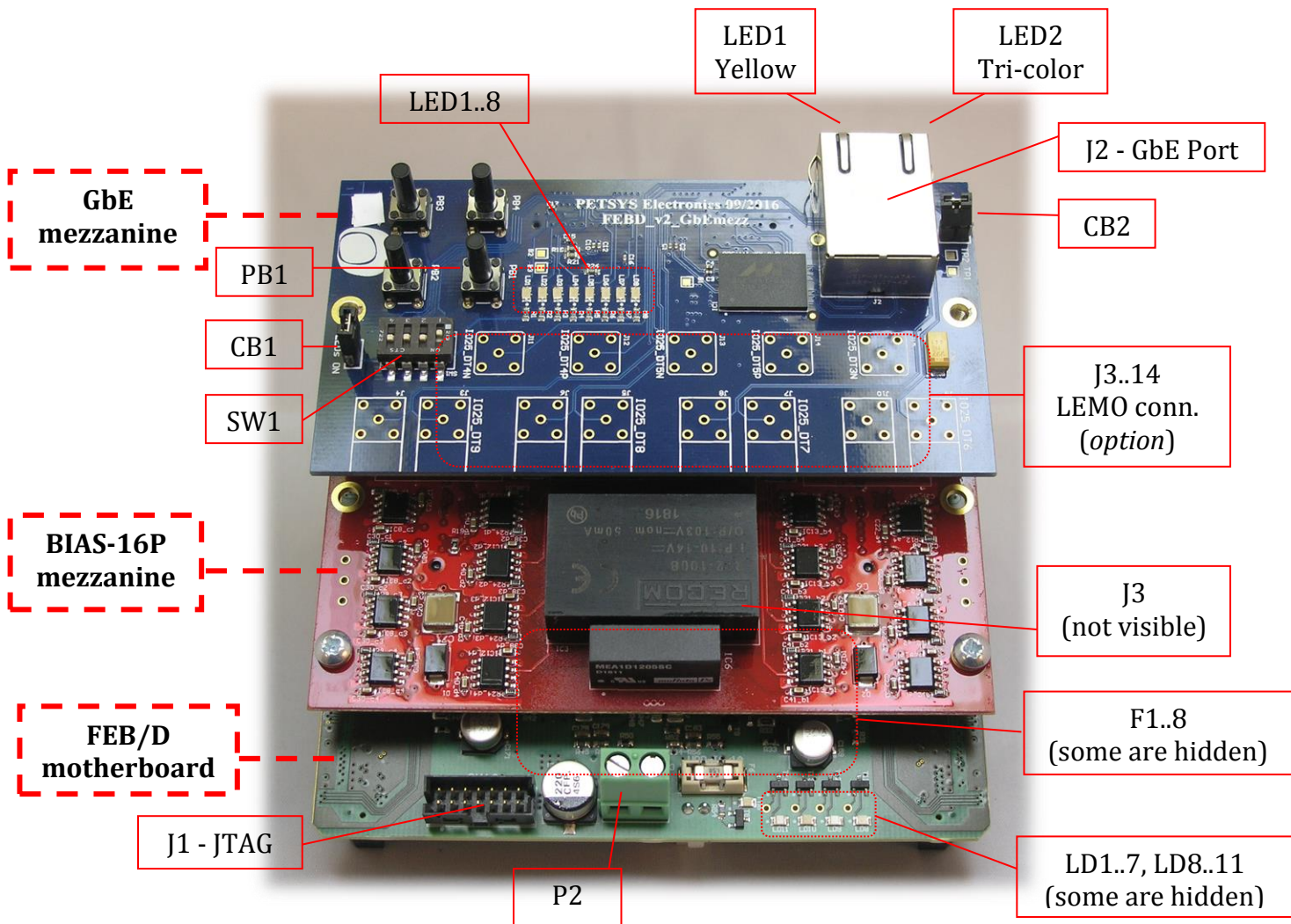


Figure 2 – Main components on top side of FEB/D_v2 set

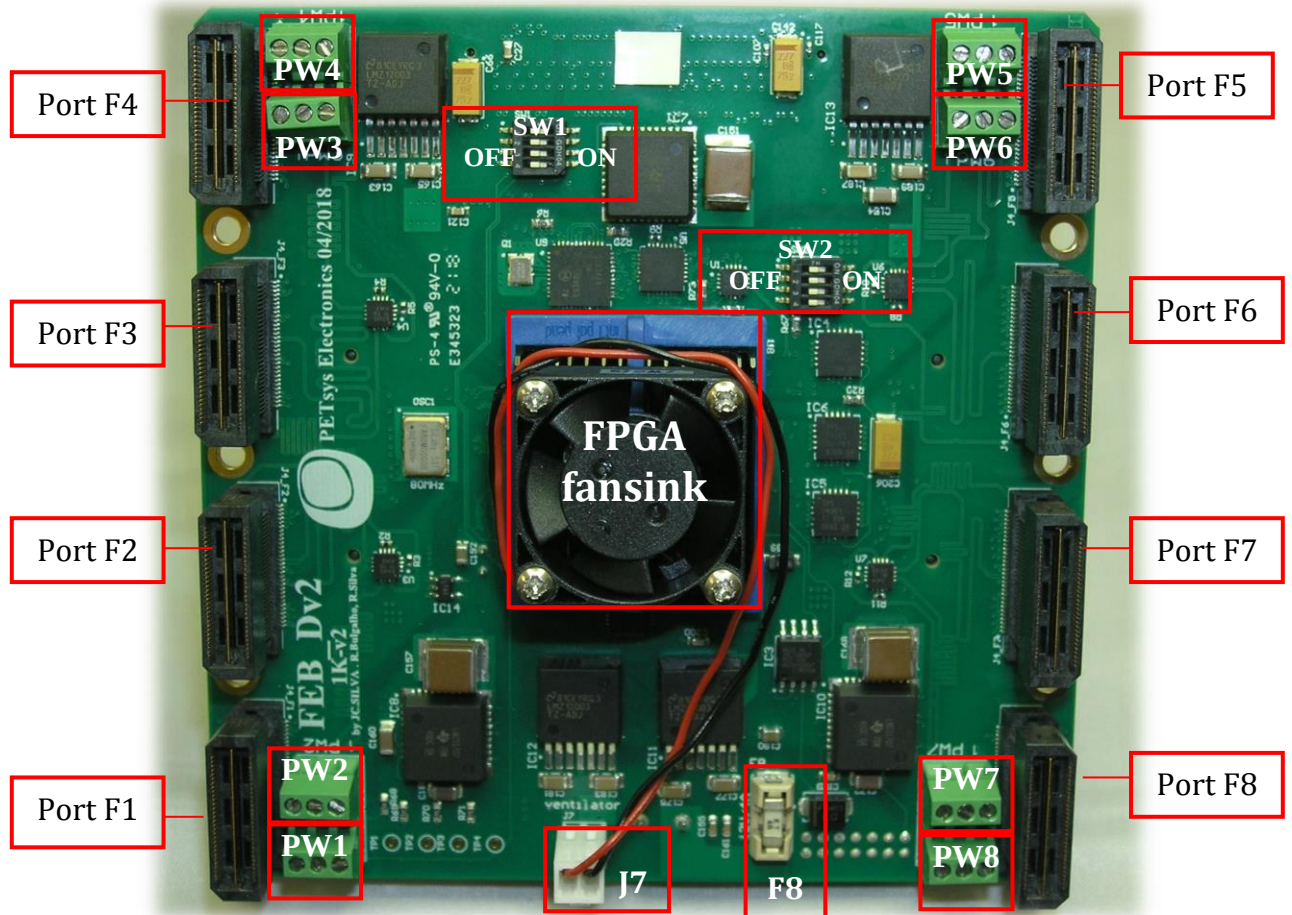


Figure 3 – Main components on bottom side of FEB/D_v2 set

4.1.1 Motherboard

P2 – Power supply input (PHOENIX_1715721)

The nominal input voltage should stay within $12 \pm 10\%$ Vdc, being the nominal/maximum input current 3/4 A when fully populated with FEMs.

The following picture highlights the polarity of the power supply connector:

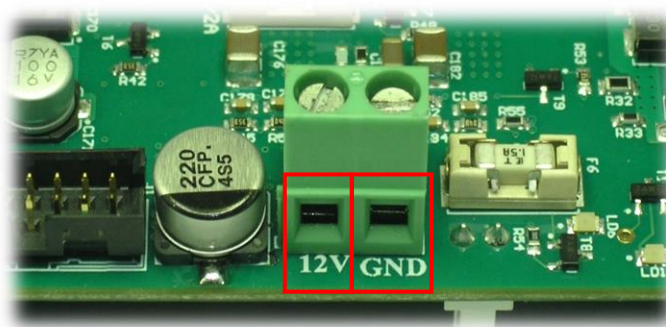


Figure 4 – Power supply input polarity

(!) Important notes:

The picture shows the polarity of the power supply connector in the FEB/D_v2-1Kv2 motherboard, which is swapped relative to the previous motherboard version, namely FEB/D_v2.

The absolute maximum input voltage is 14 Vdc, beyond which may severely damage the board.

F1..F8 – FEB/D_v2 has 8 protecting fuses. If needed please consult “TOF FEB-D_v2 - Hardware User Guide” manual to get specific information.

J1 – JTAG Interface (see Firmware section)

Port F1..F8 – FEM ports (SAMTEC_QTH-030-04-L-D-A)

FEB/D_v2 has a total of 8 FEM ports available. Each FEM port provides power, clock signal, slow control lines, return path for each ASIC data output link and reads the temperature sensors on the FEM. Please read “PETsys TOF FEB-D_v2 - Hardware User Guide” to get the pin-out of these connectors.

PW1..8 – FEM extra power supply connector (TYCO_282834-3)³

By default, the FEB/D_v2 is able to supply a FEM located at any distance up to 50 cm away by means of an [HQCD](#) series flat-cable from SAMTEC. However, for longer distances, these extra connectors (FEB/D_v2-1Kv2 feature only) must also be used. Please read also “*PETsys TOF FEB-D_v2 - Hardware User Guide*” to get further information.

J7 – FPGA cooling fan (MOLEX_22-27-2021)

FEB/D_v2 is shipped with a fan plus an heatsink to dissipate the heat from the FPGA. This connector provides 12 Vdc to drive such fan.

(!) Important note: *Unless strictly necessary, DO NOT remove this fan, nor unplug it from connector J7.*

J3 – LEDs blanking function

If needed, remove this jumper to switch OFF all LEDs of the motherboard (LD1..11), preventing this way any light contamination to the measurements with SiPMs.

(!) Important note: *To change the default ON state of this jumper it is necessary to disassemble the GbE mezzanine and the BIAS mezzanine as well. Please, see BIAS-16P Mezzanine section.*

LD1..7 – These LEDs show the status of the different power rails of the FEB/D_v2. If needed please consult “*TOF FEB-D_v2 - Hardware User Guide*” manual to get specific information.

LD8..11 – These LEDs have no function assigned yet.

SW1, SW2 – Temp. sensors (1-wire bus by-pass switch)

These switches only apply to the former TOFPET2 ASIC Test Boards.

³ Feature only available in the FEB/D_v2-1Kv2 motherboard

4.1.2 BIAS-16P Mezzanine

FEB/D_v2 supplies bias voltage to the SiPMs through this mezzanine. This mezzanine provides 16 positive bias voltages that can be independently set within the range 0..75 V, and can supply 20 mA peak, or 2.5 mA on average per bias line (default mezzanine type). Please, read also carefully “TOF FEB-D_v2 - Hardware User Guide”, which gives further details about this mezzanine board. A customised mezzanine version, e. g. supplying negative bias voltage, or with larger current, can be developed on request.

(!) Important note: *If for some reason it needs to be re-assembled it MUST solely be inserted on the motherboard exactly as shown by the following photo, i.e., J6 and J5 connectors MUST mate respectively with connectors J5 and J6 of the motherboard, otherwise fuse F8 (see Figure 3) will either degrade or blow up!*

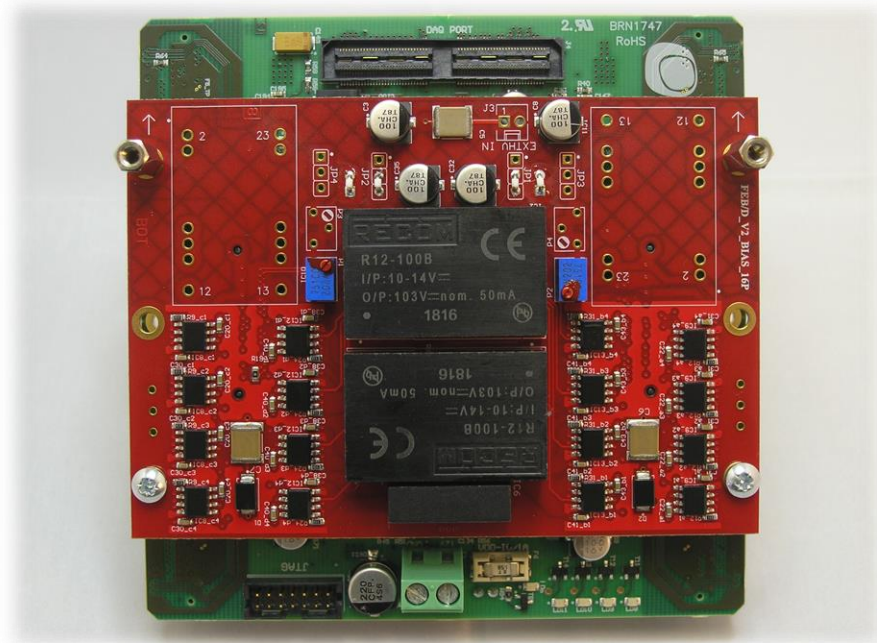


Figure 5 – Photo showing the correct way to assemble the BIAS-16P mezzanine

4.1.3 GbE Mezzanine

The Giga Bit Ethernet mezzanine is the main interface to the computer and provides the access and control of the Evaluation Kit over the ethernet port.

The Ethernet connector meets the 1000BaseT standard (also known as IEEE 802.3ab) meaning that is able to handle 1 Gbit/s over network cables up to a maximum length of 100 m, provided that a cat.5 twisted pair cable (or better) is used to communicate with the Ethernet port of the data acquisition computer.

Meaning of LED colors:

LED1 – Yellow when the Ethernet link is established

LED2 – Green when transmitting packets, Orange when receiving packets, or Yellow when there is activity in both ways.

J2 – RJ45 Gigabit Ethernet connector (BEL FUSE_L829-1J1T-43)

J3..J14 – (Option) Coaxial 50 ohm connectors (LEMO_EP-L/A-.00.250.NTN)

These LEMO connectors are available (*under customer's request*) to meet any further test needs (*no function assigned yet by firmware*).

CB1 – Remove this jumper to switch off the light issued by LED1..8

CB2 – Remove this jumper to switch off the LEDs of the Ethernet port (J2)

SW1 – This 4-way DIP switch is available to meet any further test needs (*no function assigned yet by firmware*).

LED1..8 – These LEDs are available to meet any further test needs (*no function assigned yet by firmware*).

PB1 – RESET button

Press this button if for some unknown reason the Ethernet link is not successfully established.

PB2..PB4 – These push-buttons are available to meet any further test needs (*no function assigned yet by firmware*).

4.2 Front-End-Module

The standard Front-End-Module (FEM-128) is made up of one interface board called FEB/I, plus two FEB/A_v2 boards (each with one 64-ch. High Performance TOFPET2 ASIC), and one FEB/S board able to host two 8x8 SiPM arrays, making a set of 128 readable SiPM channels. Up to eight PETsys Front-End Modules can be connected to one FEB/D board either directly, or using an [HQCD](#) series cable from SAMTEC.

(!) Important note: We strongly recommend you to read carefully the Front-End-Module section of “PETsys TOFPET2 ASIC SiPM Readout System - Hardware User Guide”, to get complete information about the FEB/A, FEB/I and FEB/S boards before proceeding with the reading of this manual.

The following photo displays the FEM⁴ in close details:

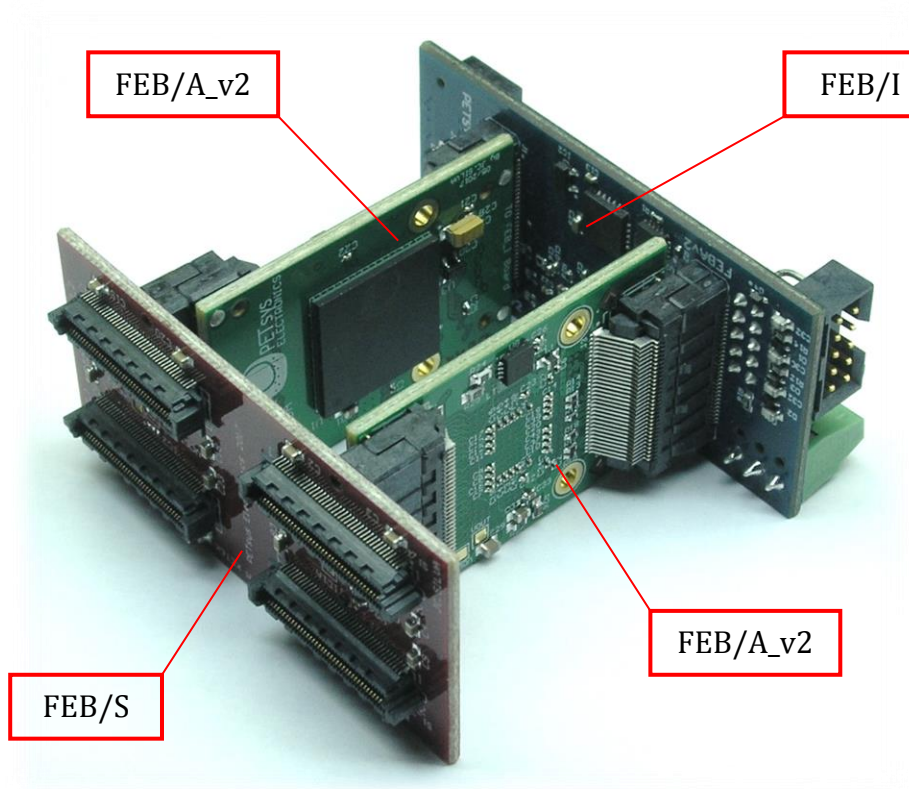


Figure 6 – Photo of the Front-End-Module (FEM)

⁴ The FEM-128 has two FEB/A_v2 boards whereas the FEM-64 has one FEB/A_v2 board and one dummy FEB/A_v2 board for mechanical stability

The Evaluation kit is shipped by default with two basic Front-End-Modules (FEM-64), that is, each FEM has only one FEB/A_v2 board with TOFPET2 ASIC (besides a dummy FEB/A for mechanical stability).

Also, each FEM-64 is shipped by default with one FEB/S board⁵, called FEB/S-0808-SS4-27.5, which is able to host two 8x8 SiPM arrays compatible with KETEK (e.g. p/n PA3325-WB-B0-A0-0808) and HAMAMATSU (e.g. p/n S13361-3050AE-08) manufacturers. However, since the FEM-64 only has one FEB/A_v2, it will read only the corresponding 8x8 SiPM array.

The following photos show both sides of the FEB/S-0808-SS4-27.5 in close details:

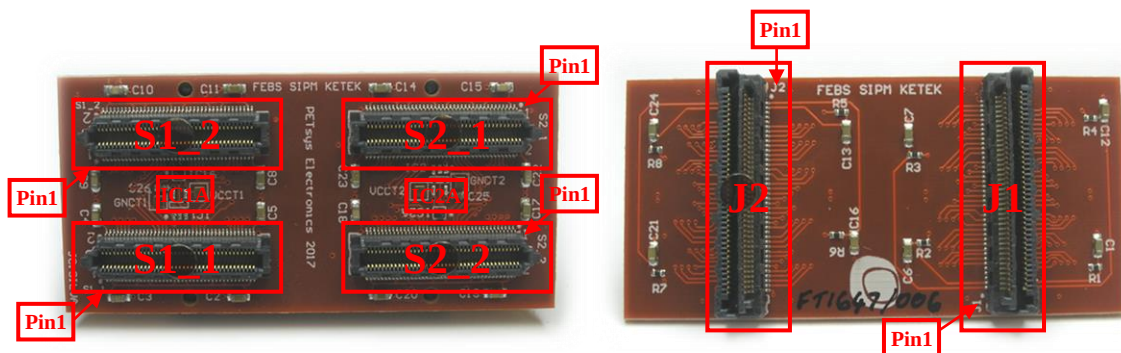


Figure 7 – FEB/S-0808-SS4-27.5 board: (left) front view, (right) back view

IC1A, IC2A – High-precision temperature sensors (2x TI_LMT70YFQR)

The FEB/S-0808-SS4-27.5 has two high-precision temperature sensors, each facing the bottom side of the SiPM array once inserted.

J1, J2 – FEB/A_v2 connectors (2x SAMTEC_LSHM-140-04.0-L-DV-A-N)

These connectors interface with FEB/A_v2. Please read “*PETsys TOFPET2 ASIC SiPM Readout System - Hardware User Guide*” to get the pin-out of these connectors.

S1_1, S1_2 – Connectors for 8x8 SiPM array 1 (2x SAMTEC_SS4-40-3.00-L-D-K-TR)

⁵ The customer has also the option to design his own FEB/S suitable for the SiPM arrays of his own choice from any vendor

S2_1, S2_2 – Connectors for 8x8 SiPM array 2 (2x SAMTEC_SS4-40-3.00-L-D-K-TR)

These four keyed connectors are compatible with 8x8 SiPM array from KETEK (e.g. p/n PA3325-WB-B0-A0-0808) and HAMAMATSU (e.g. p/n S13361-3050AE-08). Please read “PETsys TOFPET2 ASIC SiPM Readout System - Hardware User Guide” to get the pin-out of these connectors.

(I) Important notes:

- a) *SiPM cathode pins are biased positively with regard to ground.*
- b) *Each array is fully biased with a single biasing line coming from the BIAS-16P mezzanine board.*
- c) *When a FEM-128 is plugged on Port F1 of FEB/D_v2 (see Figure 3) ASIC channel ranges from 0..63 (ASIC0) and 64..127 (ASIC1). When plugged on Port F2, ASIC channel ranges from 128..191 (ASIC2) and 192..255 (ASIC3). When plugged on Port F3, ASIC channel ranges from 256..319 (ASIC4) and 320..383 (ASIC5), and so on until Port F8, i.e., 896...959 (ASIC14) and 960..1023 (ASIC15).*
- d) *The lower ASIC channel range of the FEM-128 corresponds to the FEB/A connected to J1 in the FEB/I, whereas the higher range corresponds to the FEB/A connected to J2 in the FEB/I.*
- e) *In the case of the FEM-64, the ASIC range will as well depend on which connector of the FEB/I the single FEB/A is plugged in. Also, it will only read the corresponding 8x8 SiPM array, that is, if the FEB/A is attached to connector J1 of FEB/S it will read the SiPM array plugged in both S1 connectors, or else if the FEB/A is attached to J2 of FEB/S it will read the SiPM array plugged in both S2 connectors.*
- f) *The FEB/D ports are split in 4 coincidence trigger regions (F1&F2, F3&F4, F5&F6, F7&F8 ports). Coincidences are only possible between different trigger regions. Please read "PETsys TOFPET2 ASIC Evaluation kit - Software User Guide".*
- g) *For simplicity we recommend to plug one of the FEM-64 shipped with the Evaluation kit on Port F1 and the second FEM-64 on Port F3 in order to obtain coincidences.*

Additionally, two 50 cm long flat flexible cables (SAMTEC [HQCD-030-20.00-TTL-SBL-1-N](#)) are also supplied to be installed between the FEB/D_v2 and the FEMs, allowing to carry out measurements in the mechanical setup shipped with the Evaluation kit.



Figure 8 – HQCD-030-20.00-TTL-SBL-1N from SAMTEC

Furthermore, for an easy start, the Evaluation kit also includes two small SiPM adapters, each with one KETEK SiPM pixel glued with one LYSO crystal of 3x3x5 mm² wrapped with Teflon that fit in any of the four connectors in the FEB/S, and a source holder for placing a (1 inch disc diameter) ²²Na point source (*note: radioactive source not supplied*), as can be seen in Figure 9, which is 4-position configurable allowing making coincidence measurements with different pairs of channels.

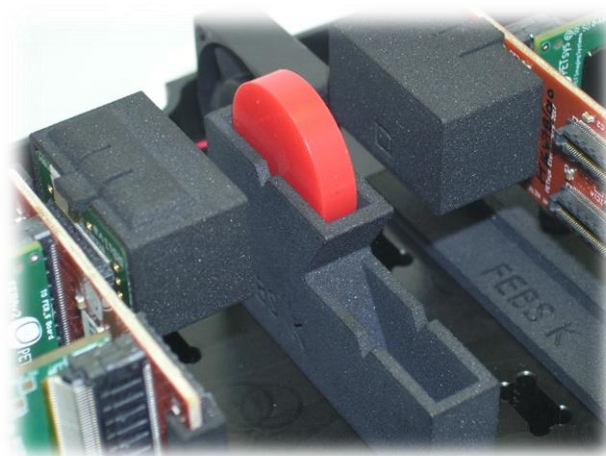


Figure 9 – Radioactive source holder aligned between SiPM pixels

4.3 Cooling system

PETsys Evaluation kit comprises a mechanical setup that provides a dark and cooled environment required to carry out stable and reproducible measurements with SiPMs.

This mechanical setup consists of a black aluminium box, thermally insulated, and cooled with peltier elements (whose temperature regulation can be done either manually or automatically), sitting on a tailored base plate with cable passages, two FEM holders (shipped already fastened to the base plate in the recommended configuration) plus the radioactive source holder, and a small 12 Vdc fan (shipped aside in a plastic bag along with the source holder) to keep the air circulating inside.

The base plate has one ‘drawer’ on each side that needs to be pulled away to reveal a passage through which the cables circulate from outside to inside. Don’t forget to put back the drawer.



Figure 10 – Base plate drawer

Place the small 12 Vdc fan in its dedicated position on the base plate and connect it to the 2-wire cable coming from the cooling box (use red/red and black/black colour rule), as the following picture shows.

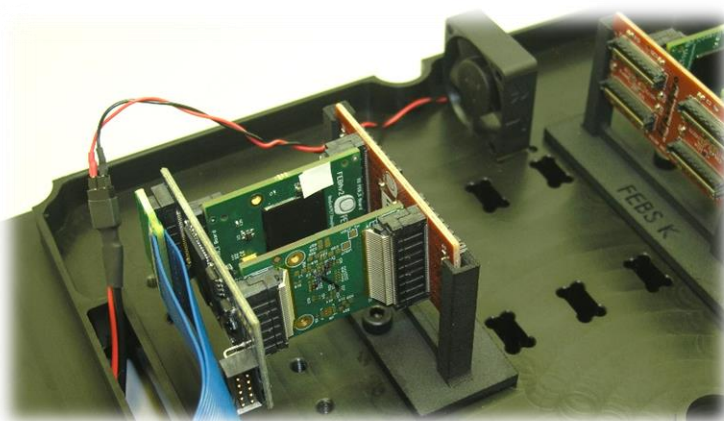


Figure 11 – Location and wiring of the small 12 V fan in the base plate

When we ship the two FEM-64, one FEM-64 differs from the other by the chosen connector of the FEB/I and the FEB/S where the single FEB/A board is plugged in. This configuration allows getting coincidences when both FEM-64 are assembled in the base plate. To ensure this, slide down the two FEM-64 in their holders in such a way that both FEB/A v2 are aligned to each other (see dotted line in Figure 12). This will in turn ensure that the provided SiPM Ketek pixel adapters will then make coincidence with the point source placed in-between with the help of its holder, which can be flipped allowing to choose two alignment heights, depending on the chosen connectors where the SiPM pixels adapters are plugged in the FEB/S.

The following two photos show the two alignment heights between SiPM Ketek pixel adapters:

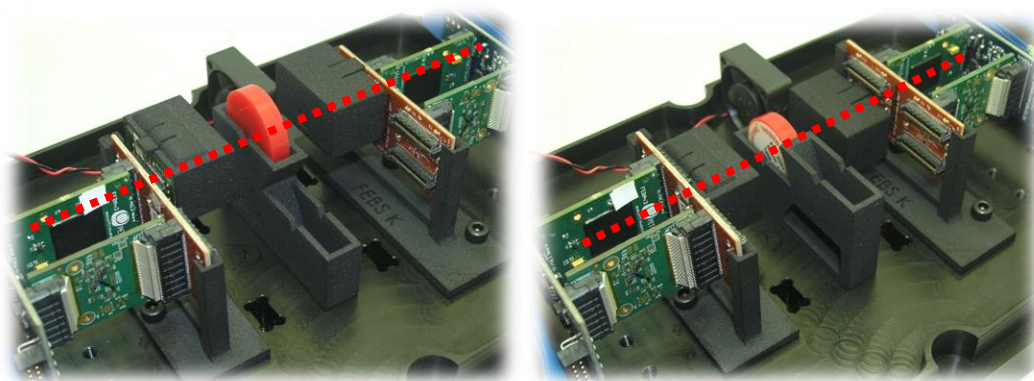


Figure 12 – Two alignment heights between SiPM Ketek pixel adapters

One of the FEM-64 is shipped with the FEB/A plugged between connector J1 of FEB/I and connector J1 of FEB/S, and the other FEM-64 is shipped with the FEB/A plugged between connectors J2 respectively. The corresponding ASIC channel number as a function of the chosen connectors in the FEB/S where the SiPM Ketek pixel adapter is plugged in, the FEB/A type shipped (COB or BGA), and the FEB/D port where the FEM is plugged in (recommended ports are F1 and F3 – see points f) and g)), is shown in the following table:

FEB/A	FEB/S	FEB/A_COB		FEB/A_BGA	
		Port F1	Port F3	Port F1	Port F3
J1	S1_1	8	264	20	276
	S1_2	33	289	34	290
J2	S2_1	72	328	84	340
	S2_2	97	353	98	354

Table 1 – ASIC channel versus SiPM Ketek pixel adapter location

Connect one of the provided HQCD cables to Port F1 in the FEB/D and the other cable to Port F3 (as recommended in point g). Then pass the HQCD cables through the side entries of the base plate, and before plugging them to the FEM, fold them exactly as shown by the following photo:

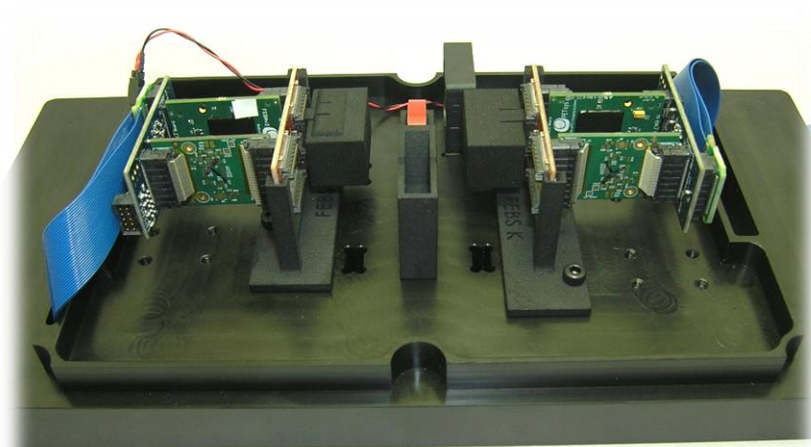


Figure 13 – Detail of the HQCD cables in the base plate

Cover the setup with the cooling box, as shows the following photo:

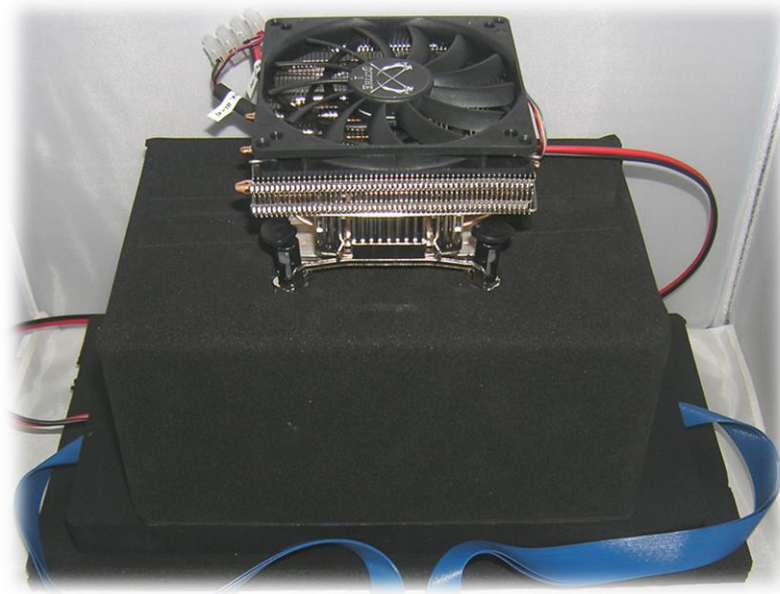


Figure 14 – Cooling box

There are two Peltier elements (p/n ET-161-12-16-E) electrically connected in series that require a maximum of 30 V and 4 A to cool the box and extract the small amount of heat produced by the ASICs. Temperature control inside the box can be either manual (approximative), using an adjustable power supply, or automatic (recommended) using a temperature control device:

- 1) **Manually** – Connect the terminal block (see Figure 15) to an external power supply, adjustable up to 30 V and able to deliver 4 A. Then manually adjust the voltage aiming at a desired temperature (e.g. 19 °C). FEB/S sensor temperature can be used for this purpose.
- 2) **Automatically** – For precise temperature control a TE controller is recommended like TC-48-20, or FTC100D, with a dedicated 30 Vdc/4 A power supply.

Once started, the system takes about 15 minutes to stabilize temperature.

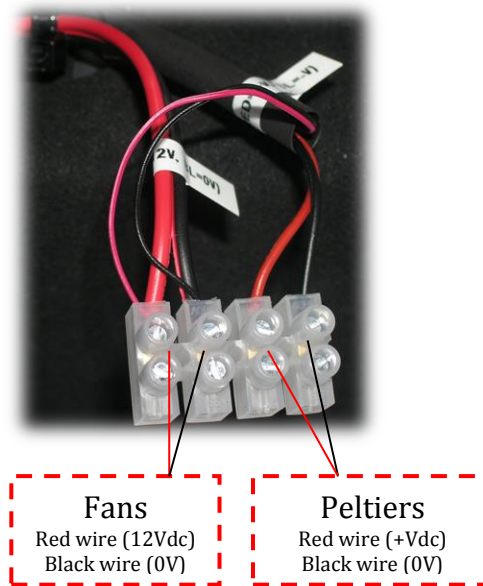


Figure 15 – Terminal block connections (Fans and Peltiers)

(!) Important notes:

- FAN must be ON when PELTIER elements are ON. The heat produced by the peltier elements, if not extracted, will dominate over the cold side and eventually damage the cooling box. Connect a 12Vdc/0.4 A to the terminal block, as indicated by Figure 15.*
- DO NOT leave the cooling system ON for a long period, like overnight, due to condensation inside the box.*
- Use of silica bag inside the box is recommended.*

5 Firmware

The FEB/D_v2 board contains a standard JTAG interface (see connector J1 on Figure 2) for direct connection with a programming cable, such as Xilinx Platform cable USB. The JTAG Interface can be used for programming purposes, either to load a different firmware version into the Flash Memory, which comes already programmed by default, or directly into the Kintex 7 FPGA_XC7K160T, or it can be used for debugging purposes. The firmware files are distributed when purchasing the hardware.

If there is a need to change the firmware please follow the instructions given under section *Firmware* of “*TOF FEB-D_v2 - Hardware User Guide*”

(!) Important note: *The JTAG programmer must be first removed before powering up FEB/D_v2, otherwise the FPGA may not read properly the content (firmware) of the flash memory.*

Contact support@petsyselectronics.com in case of problems.

6 Software

Please read *"PETsys TOPPET2 ASIC Evaluation kit - Software User Guide"*

Contact support@petsyselectronics.com in case of problems.

7 Physical Dimensions

7.1 FEB/D_v2 set

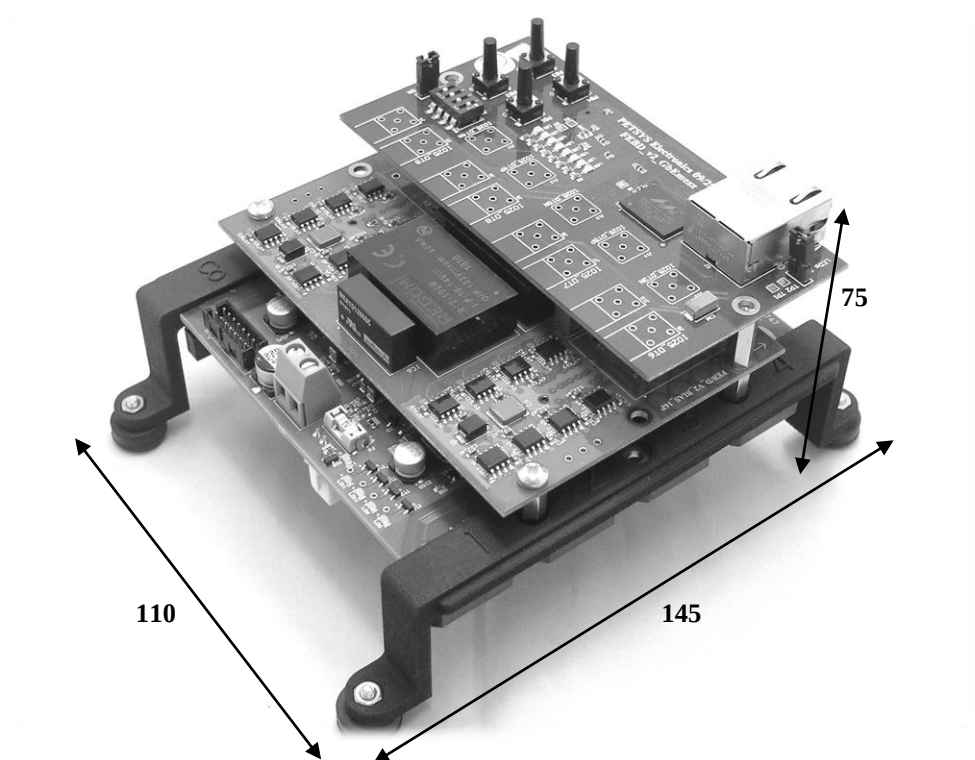


Figure 16 – FEB/D_v2 set dimensions (mm)

7.2 FEM

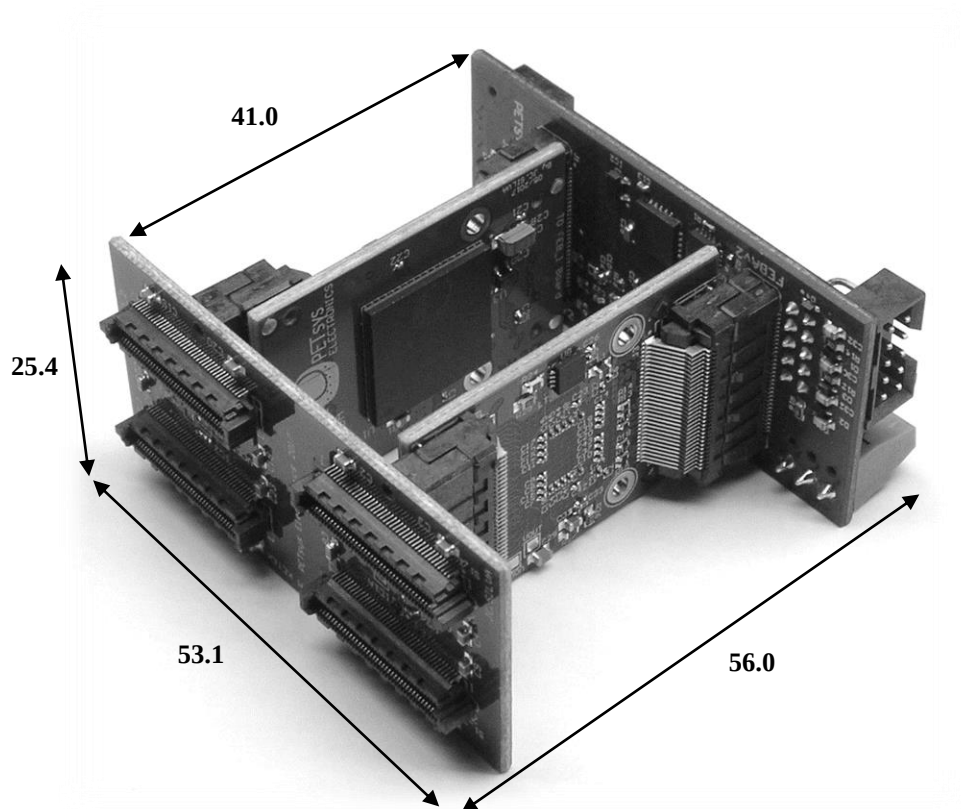


Figure 17 – FEM overall dimensions (mm)

7.3 Mechanical setup

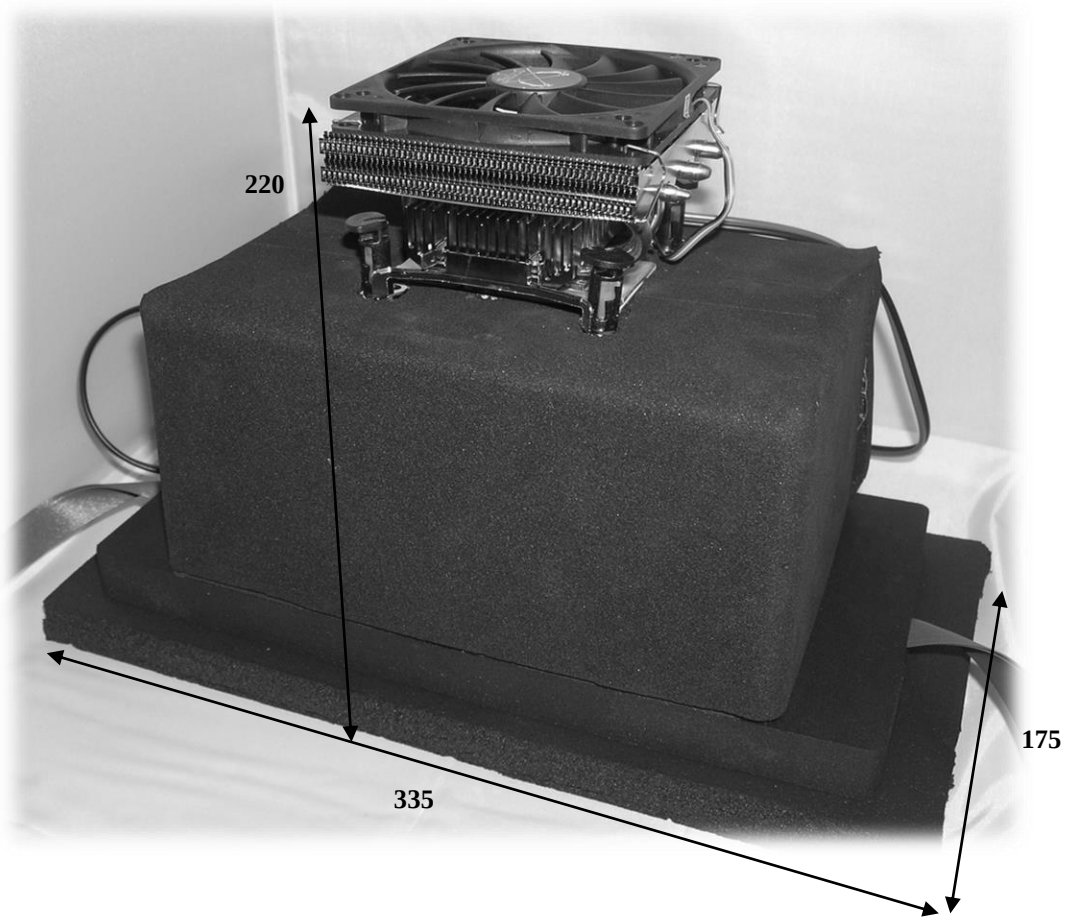


Figure 18 – Mechanical setup dimensions (mm)

8 Disclaimer

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